

AWS EC2: Elastic Compute Cloud

A Professional Overview

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Batch: P526

Course: Executive program on DevOps with AWS

What is Amazon EC2?

Amazon Elastic Compute Cloud (EC2) is a web service that provides secure, resizable compute capacity in the cloud. It is designed to make web-scale cloud computing easier for developers.

Key Benefits:

- **Elasticity:** Scale capacity up or down in minutes.
- **Control:** Complete control over your instances (root access).
- **Flexibility:** Choice of multiple instance types, operating systems, and software packages.
- **Cost-Effective:** Pay only for the capacity you actually use.

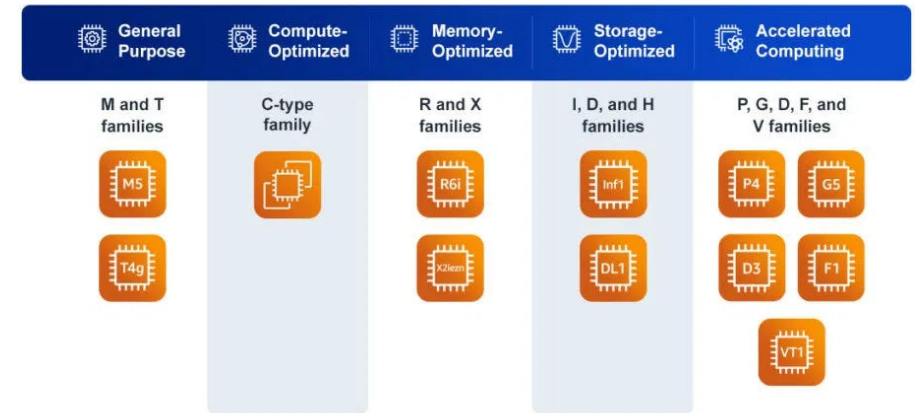
Core Features of EC2

- **Instances:** Virtual computing environments.
- **Amazon Machine Images (AMIs):** Preconfigured templates for your instances.
- **Instance Types:** Various configurations of CPU, memory, storage, and networking capacity.
- **Key Pairs:** Secure login information for your instances.
- **Instance Store Volumes:** Temporary storage for data that is deleted when you stop or terminate your instance.
- **Amazon EBS Volumes:** Persistent storage volumes for your data.
- **Security Groups:** A virtual firewall that controls traffic for your instance.

EC2 Instance Families

- **General Purpose:** Balanced compute, memory, and networking for diverse workloads (e.g., T4g, M6g).
- **Compute Optimized:** Ideal for compute-bound applications like high-performance web servers (e.g., C6g, C7g).
- **Memory Optimized:** Designed for fast performance for workloads that process large data sets in memory (e.g., R6g).
- **Storage Optimized:** High, sequential read/write access to very large data sets on local storage (e.g., I4i, D3).
- **Accelerated Computing:** Uses hardware accelerators for functions like graphics or data processing (e.g., P4, G5).

Different AWS EC2 Instance Types



EC2 Pricing Models

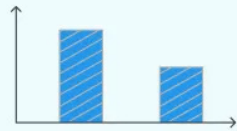
EC2 Instance Pricing Models



On-Demand



Reserved Instance



Spot Instance



Dedicated Host

- **On-Demand:** Pay for compute capacity by the second or hour with no long-term commitments. Ideal for short-term, irregular workloads.
- **Savings Plans:** Reduce costs by committing to a consistent amount of usage (measured in \$/hour) for a 1 or 3-year term.
- **Reserved Instances:** Provide a significant discount (up to 72%) compared to On-Demand pricing for a committed 1 or 3-year term.
- **Spot Instances:** Request unused EC2 capacity at steep discounts (up to 90% off). Best for fault-tolerant and flexible applications.
- **Dedicated Hosts:** A physical EC2 server dedicated for your use, helping you meet regulatory requirements and reduce costs by using existing server-bound software licenses.

EC2 Networking & Security

- 🛡️ **Virtual Private Cloud (VPC):** Provision a logically isolated section of the AWS Cloud to launch resources in a defined virtual network.
- 🛡️ **Elastic IP Addresses:** Static IPv4 addresses designed for dynamic cloud computing, allowing you to mask instance failures by remapping addresses.
- 🛡️ **Security Groups:** Act as a virtual firewall for your instance to control inbound and outbound traffic at the instance level.
- 🛡️ **Network ACLs:** An optional layer of security for your VPC that acts as a firewall for controlling traffic in and out of one or more subnets.
- 🛡️ **IAM Roles:** Securely delegate access to AWS services and resources for your EC2 instances without using long-term security credentials.

Amazon Machine Images (AMI)

An AMI provides the information required to launch an instance, acting as a master image for the virtual server.

Core Components of an AMI

- **Root Volume Template:** Contains the Operating System, application server, and applications.
- **Launch Permissions:** Controls which specific AWS accounts are authorized to use the AMI.
- **Block Device Mapping:** Specifies the volumes to attach to the instance upon launch.

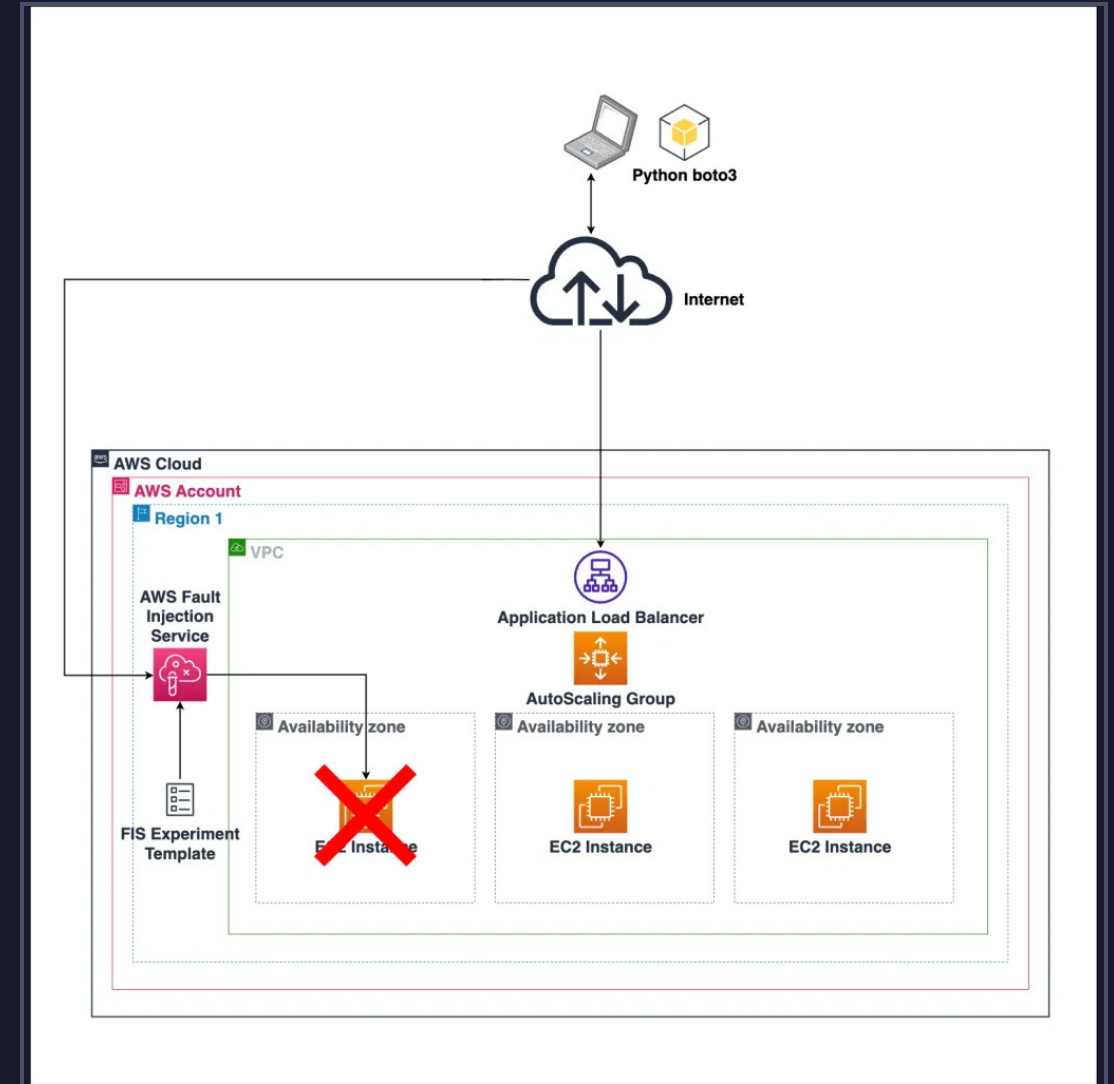
Available AMI Types

- **AWS Marketplace AMIs:** Managed by third-party vendors for specific software stacks.
- **Community AMIs:** Shared by other AWS users for general use.
- **Custom AMIs:** Created by users to capture specific configurations and software.

Scaling with Auto Scaling

Amazon EC2 Auto Scaling helps you maintain application availability by automatically adding or removing EC2 instances according to conditions you define.

- **Fleet Management:** Automatically detects and replaces unhealthy instances to maintain your desired capacity.
- **Scheduled Scaling:** Scale your infrastructure based on known traffic patterns or predictable load changes.
- **Dynamic Scaling:** Scale in real-time based on metrics like CPU utilization, memory usage, or network traffic.



EC2 Best Practices



Security First

Use IAM roles instead of sharing credentials; keep your AMIs updated and use Security Groups for least-privilege access.



High Reliability

Deploy instances across multiple Availability Zones (AZs) to ensure application availability during failures.



Proactive Monitoring

Use Amazon CloudWatch to monitor metrics and set alarms for automated responses to performance issues.



Cost Optimization

Use Spot instances for fault-tolerant workloads and Savings Plans or Reserved Instances for predictable usage.



Performance Tuning

Choose the right instance type for your workload; utilize Enhanced Networking and EBS-optimized instances.

Amazon EC2 is a foundational service of AWS, providing the "compute" in the cloud. Its versatility, scalability, and integration with other AWS services make it a primary choice for modern application deployment. Understanding EC2 is essential for any DevOps professional working with AWS.

Thank You!

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